

Andrew W. Shenstone

Shenstone, Andrew - Application for Senior Software Engineer, Data Infrastructure @ ApartmentIQ

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SUMMARY

Senior software engineer with 12+ years shipping production systems where data and integrations are the product. I am applying to help evolve ApartmentIQ's shared data platform: ingestion quality, pipeline reliability, and datasets that unlock customer-facing features across Market Research, Market Surveys, and Daylight Revenue.

I bring roughly six years of **Ruby on Rails** in production, **ETL-style pipelines** at Tune, complex **PostgreSQL** reporting and modeling at Bloom and Plantfolio, and async ingestion in production today (webhooks, background jobs). I work autonomously from concept through deployment, partner with product and application engineering on roadmap trade-offs, and communicate outcomes clearly to stakeholders.

Python, web scraping at scale, and dbt/Airflow are not my headline tenure; I have shipped reliable batch-style and event-driven workflows and ramp quickly on AWS beyond S3/EC2. Multifamily rentals is new domain; B2B data and martech-shaped products are not.

SKILLS

Data platform & pipelines: ETL-style pipelines and marketplace integrations in production (Tune Inc), Async ingestion patterns: webhooks, background jobs, and queue-like workflows (Inngest on Plantfolio), Structured feed and API integration across commerce, CMS, and payments vendors (Bloom, Plantfolio), PostgreSQL: complex reporting queries, relational modeling, migrations, and performance-aware SQL (Bloom, Plantfolio/Prisma)

Rails, backend & product delivery: Ruby on Rails (~6 years through Rails 6): GraphQL APIs, monolith patterns, and operator-facing features (Bloom; agency Rails work), Node.js services and API design; JSON/XML-shaped integrations at the application layer, End-to-end ownership from scoping with product through deployment, monitoring, and iteration, Documentation, code review, and mentoring habits that match platform knowledge-sharing expectations

Cloud, observability & stack ramps: AWS in production (S3, EC2); RDS, ECS, Lambda, and OpenSearch align to your stack with a focused ramp, CI/CD (CircleCI, Vercel/Turbo); testing discipline (Vitest, Cypress) on paths where bad data breaks trust, Python, web scraping at scale, dbt, and Airflow are not headline tenure; batch-style and event-driven workflows in Node/Rails are, B2B SaaS and martech-adjacent delivery (Bloom integrations; data-heavy admin and reporting surfaces)

AI-native product & workflow: AI-native product delivery: vision, conversational UX, RAG/embeddings in production (Plantfolio), OpenAI API, AI SDK, vector embeddings, AI-assisted engineering: Cursor (top 1% user), Claude Code—class workflows—daily practice since the current wave of coding agents, not a recent experiment, Codebases structured for agent-friendly work: typed TypeScript monorepos, tests, clear package boundaries

EDUCATION

B.S. Finance, Santa Clara University — 2010

EXPERIENCE

Founder & Lead Engineer

Plantfolio | November 2023 - Present

AI-native product & engineering

- Built Plantfolio as an AI-native application: plant ID from photos, contextual assistant (Sage), smart tag extraction, and semantic search over 40,000+ species—LLM features customers use in production, not demos

- AI-native way of building: Cursor (top 1% user) and Claude-class agents are my default workflow from early in the product; I own architecture and review while using AI for speed across the 12+ package monorepo
- Designed the monorepo for agent-friendly iteration—TypeScript, clear boundaries, Vitest/Cypress, and consistent patterns so AI-assisted shipping stays maintainable

Product Leadership & Architecture

- Architected and built a comprehensive plant management platform from the ground up, serving as both technical lead and product owner
- Owned the full feature cycle: joined customer calls to identify friction points, architected solutions, implemented, tested, and verified success in production
- Led product strategy and user experience decisions, making product micro-decisions and navigating ambiguity to deliver value iteratively

Technical Achievements

- Modern Frontend Architecture: Implemented Next.js with React, React Server Components, and TypeScript, achieving exceptional performance and developer experience
- Data modeling & migration: Built Plantfolio on MongoDB and shipped production there for 1+ year, then migrated late to PostgreSQL on Neon (15+ interconnected models with Prisma today)—deep familiarity with document and relational modeling tradeoffs
- Database choices: Evaluated Supabase during exploration but did not ship on it; chose Neon for managed Postgres; auth and subscription tiers via Clerk with Postgres-centric access patterns in production today
- Content Management: Created sophisticated content management system with rich text editing, plant tagging, and flexible data structures

Scale & Performance

- Managed a database of 40,000+ plant species with advanced search capabilities using Algolia and vector embeddings

- Implemented comprehensive media management system with AWS S3, supporting image optimization, galleries, and metadata extraction
- Built robust authentication and authorization systems with Clerk, supporting multiple user types and subscription tiers
- Optimized application performance through React Server Components, efficient data fetching, and strategic caching

Client Case Study - AI Signage

- Delivered a production rollout for an independent garden center to automate plant signage generation; one independent retail customer in ongoing production who has been an enthusiastic reference
- Enabled bulk plant import (plain text) to printable signs in seconds; reduced a weekend-long manual process to minutes after initial setup
- Demonstrated ability to translate customer requirements into scalable technical solutions and shorten feedback loops

Full-Stack Ownership

- Developed complete payment integration with Stripe, subscription management, and user onboarding flows
- Built comprehensive email system with React Email, supporting transactional emails, notifications, and marketing campaigns
- Implemented advanced search and filtering capabilities with Algolia, supporting complex plant queries and user preferences

Code Quality & Leadership

- Established coding standards and architectural patterns that enabled rapid, consistent development—with AI-assisted velocity grounded in tests and human review
- Implemented comprehensive testing strategy with Vitest and Cypress, maintaining high code quality

- Built reusable component library and design system, ensuring consistent UI/UX across the platform
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Senior Software Engineer

Tilia | January 2023 - August 2023

- Partnered with cross-functional teams in product and design to revamp critical user-facing features
 - Built and shipped work in support of KYC, identity verification, and risk-adjacent tooling as part of Tilia's payments platform (tools team context)
 - Introduced and evolved a GraphQL API on the platform while consuming it from our product—hands-on with both server-side schema/resolvers and client integration
 - Spearheaded and guided a team through the successful adoption of Chakra UI, reducing bespoke components and establishing cohesive design patterns
 - Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
 - Maintained exceptional unit test coverage of over 90% for the Tools team application, alongside comprehensive end-to-end testing
 - Led the swift creation and deployment of a homepage reporting dashboard within the initial two months of joining
 - Role ended in a company-wide layoff affecting approximately one-third of employees
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Senior Software Engineer

Bloom.wine | September 2020 - October 2022

- Architected a greenfield Next.js application for the primary admin interface of the Bloom product; much of that stack remains in production across many winery clients today

- Engineered Ruby on Rails GraphQL API endpoints shared by multiple applications across mobile and web devices
 - CI/CD with CircleCI for build, test, and deployment workflows alongside the Next.js and Rails stack
 - Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
 - Collaborated with designers using Figma, establishing component library standards usable by both engineering and design
 - Integrated DatoCMS and extended application functionality through third-party API integrations (Shopify, Mailchimp, Stripe, Shipstation, ActiveCampaign); all Bloom winery clients were on Shopify Plus
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Software Engineer II

Tune Inc - Marketplace Automation | August 2019 - May 2020

- Built and operated ETL-style data pipelines and integrations supporting marketplace automation workflows
 - Automated a client workflow using a Node.js/Express API and React.js, reducing the time to complete the task from 10-15 minutes to 10-30 seconds
 - Modernized the primary legacy React.js application by moving from a custom Webpack process to react-scripts, resulting in a 50% reduction in production build size and a 60% reduction in NPM dependencies
 - Reduced development machine CPU usage from 75% to <5% by simplifying the build pipeline and dependency graph
 - Role ended in a company-wide layoff
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Software Developer

Ply Interactive Inc | June 2014 - July 2019

React Native

- Shipped multiple production React Native applications for agency clients—cross-platform mobile apps delivered end to end
- Washington State Fair: React Native app in sustained multi-year production; very high traffic during fair season, from tens of thousands of users into the hundreds of thousands cumulatively over seasons

Web & platforms

- Developed numerous web applications including Georgetownbeer.com, WashingtonWine.org, TwoWolvesWine.com, Novaerus.com, WellAir, and Plasma-air.com
- Built a standalone Ruby gem Content Management System, enabling simplified content management within Ruby on Rails projects
- Transitioned the agency to modern technologies like GraphQL, Gatsby.js, and React.js, improving delivery times for new projects
- Bloom.wine Product Development: Built extensive reporting features using complex PostgreSQL queries and interactive filtering, saving clients thousands of hours on previously manual tasks
- Engineered a flexible Wine Club membership structure for wineries, allowing for diverse client configuration and broadening the range of potential sales targets
- Communicated and enforced the standards of Shopify's Polaris React.js component library with designers, resulting in efficiencies gained from leveraging pre-built components

COMPANY REFERENCE

Plantfolio (2023-Present) - AI-native plant management platform (vision, chat, RAG search); built with AI-assisted engineering (Cursor-first) and 40,000+ species catalog - plantfolio.garden/

Tilia (2023) - All-in-one payment platform for digital economies - tilia.io/

Bloom.wine (2020-2022) - Commerce services for the alcohol industry; all clients on Shopify Plus - bloom.wine/

Tune Inc - Marketplace Automation (2019-2020) - SaaS platform for managing marketing partnerships across mobile and web - tune.com/

Ply Interactive Inc (2014-2019) - Boutique agency: web and production React Native apps (e.g. Washington State Fair at major scale); developed Bloom.wine product internally (2017-2019)